

MODULE 1: FOUNDATIONS AND THEORETICAL APPROACH

CHAPTER 1: INTRODUCTION

- **What Is AI?**
- **The Foundations of Artificial Intelligence**
- **The State of the Art**
- **Risks and Benefits of AI**

WHAT IS AI?

Artificial Intelligence (AI) is computer software or systems that can perform tasks that normally require human intelligence.

In Everyday Language

AI is a computer program that can learn, think, and make decisions similar to how humans do — without being explicitly told how to do it for every situation.

Examples:

- ChatGPT understanding questions and writing answers
- Self-driving cars navigating traffic
- Your phone recognizing faces in photos
- Game opponents that adapt to your playing style
- Gmail filtering spam automatically

Comprehensive Definition

Artificial Intelligence is a branch of computer science that designs, develops, and implements intelligent computer systems capable of:

- Perceiving their environment through sensors or data input
- Processing information using algorithms and models
- Learning from experience and data
- Reasoning logically to solve problems
- Planning and making decisions
- Acting in the environment to achieve goals
- Adapting their behavior based on feedback and changing circumstances

RATIONAL AGENTS VS HUMAN THINKING.

What is a Rational Agent?

A rational agent is an entity that always chooses the action most likely to achieve its goal, given available information. It follows a defined logic:

Perceive → Think → Act → Maximize utility

It is the ideal model used in economics, AI, and decision theory.

What is Human Thinking?

Human thinking is shaped by biology, emotion, experience, and cognitive shortcuts (heuristics). It's fast, flexible — but also full of biases and inconsistencies.

Rational Agents	Human Thinking
A rational agent is an AI system that selects the best possible action to achieve a goal based on available information.	Human thinking is the natural cognitive process involving reasoning, emotions, intuition, experience, and judgment.
Makes decisions based on logic, algorithms, and predefined goals.	Makes decisions using reasoning, emotions, intuition, ethics, and personal experiences.

Chooses actions that maximize performance or expected outcomes.	Chooses actions influenced by knowledge, beliefs, feelings, culture, and values.
Learns from data, feedback, and machine learning algorithms.	Learns through education, observation, experience, and social interactions.
Does not possess emotions or feelings.	Strongly influenced by emotions such as happiness, fear, anger, and empathy.
Produces consistent results under the same conditions.	Decisions may vary due to mood, stress, or changing circumstances.
Processes information and makes decisions very quickly.	Generally slower because of analysis, emotions, and cognitive limitations.
Highly accurate when provided with correct data and rules.	Accuracy depends on knowledge, attention, and experience; humans may make mistakes.
Limited to programmed methods or learned patterns.	Highly creative and capable of imagination, innovation, and abstract thinking.
Adapts through algorithms and retraining.	Naturally adapts to new situations using reasoning and life experiences.

COMPUTATIONAL AGENTS

A **Computational Agent** is an intelligent software or hardware system that **perceives its environment through sensors, processes the collected information using computational techniques, and performs appropriate actions through actuators to achieve a specific goal.**

In Artificial Intelligence (AI), a computational agent is designed to act **autonomously, rationally, and efficiently** by selecting the best possible action based on its perception of the environment.

A computational agent continuously interacts with its surroundings by sensing changes, making decisions, and acting accordingly. It may also improve its performance over time through learning and experience.

Components of a Computational Agent

A computational agent consists of the following main components:

1. Sensors (Perception)

Sensors collect information from the environment. They provide the agent with the data needed to understand its surroundings.

Examples: Camera, microphone, keyboard, GPS, temperature sensor.

2. Agent Program

The agent program is the brain of the computational agent. It processes the input received from the sensors, applies algorithms, reasoning, or machine learning techniques, and decides the most appropriate action.

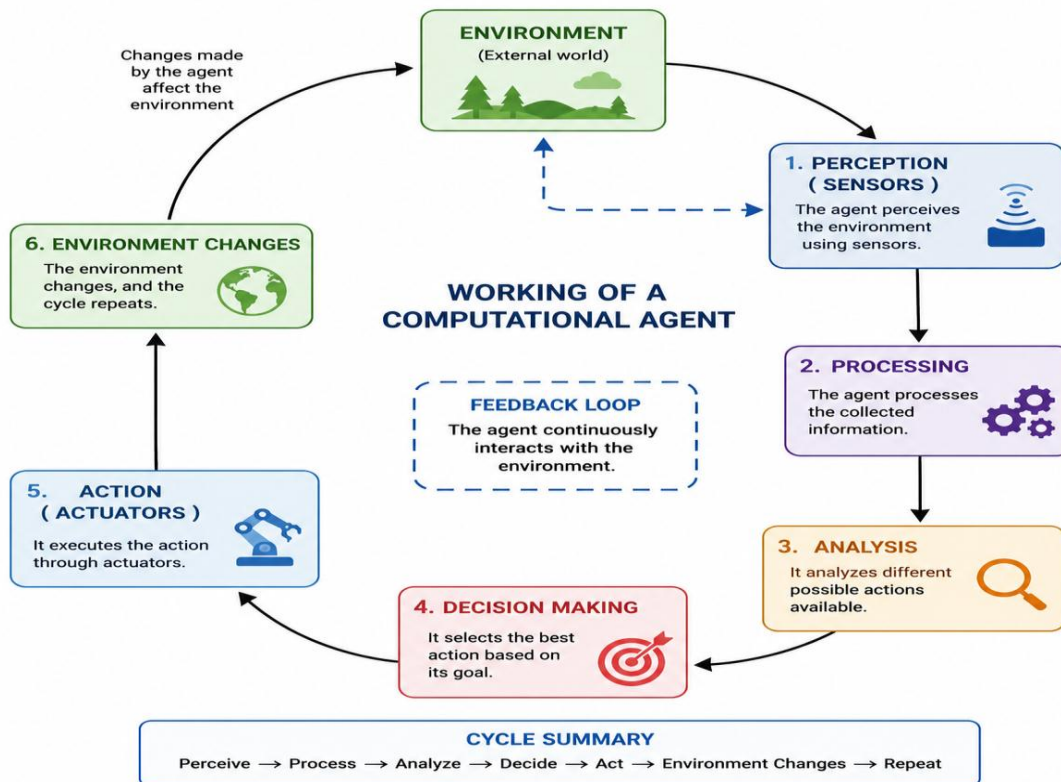
3. Actuators (Action)

Actuators execute the decisions made by the agent by performing actions in the environment.

Examples: Robot arms, display screens, speakers, wheels, motors.

4. Environment

The environment is everything surrounding the computational agent with which it interacts. It may be physical (roads, buildings) or digital (websites, databases).



AGENT-ENVIRONMENT INTERACTION,

Agent-Environment Interaction is the process by which an intelligent agent continuously **perceives its environment through sensors, processes the received information, selects the most appropriate action, and acts on the environment through actuators** to achieve a specific goal. The action performed by the agent changes the environment, creating a continuous feedback loop.

In Artificial Intelligence, an agent interacts with its environment to make intelligent decisions and improve its performance over time.

Components of Agent-Environment Interaction

1. Environment

The environment is everything external to the agent with which it interacts. It provides information (inputs) to the agent and is affected by the agent's actions.

Examples: Roads, traffic signals, websites, smart homes, game boards.

2. Agent

The agent is an intelligent system that observes the environment, processes information, and performs actions to achieve its objectives.

Examples: Robot, chatbot, self-driving car, virtual assistant.

3. Sensors

Sensors collect information from the environment and send it to the agent.

Examples: Camera, microphone, GPS, keyboard, temperature sensor.

4. Agent Program (Decision Making)

The agent program processes the information received from sensors, applies reasoning or AI algorithms, and decides the best action.

5. Actuators

Actuators execute the selected action and influence the environment.

Examples: Robot arms, motors, speakers, display screens, wheels.

TYPES OF ENVIRONMENTS.

An environment in artificial intelligence refers to the external world in which an agent operates and interacts. The behavior of an environment depends on how much information the agent can access and how the state changes over time.

- The agent perceives information from the environment using sensors and performs actions through actuators.
- Environments are categorized based on interaction style, uncertainty, and structure of decision-making.

There are eight major classifications of AI environments.

1. Fully Observable vs Partially Observable
2. Deterministic vs Stochastic
3. Competitive vs Collaborative
4. Single-Agent vs Multi-Agent
5. Static vs Dynamic
6. Discrete vs Continuous
7. Episodic vs Sequential
8. Known vs Unknown

1. Fully Observable vs Partially Observable

In a fully observable environment, the agent can access the complete state at any given time, while in a partially observable environment, only partial information is available. Fully observable systems do not require maintaining historical data, whereas partially observable systems depend on memory of past states.

Examples:

- Chess - The board is fully observable, and so are the opponent's moves.
- Driving - The environment is partially observable because what's around the corner is not known

2. Deterministic vs Stochastic

When a uniqueness in the agent's current state completely determines the next state of the agent, the environment is said to be deterministic. The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.

Examples:

- Chess - There would be only a few possible moves for a chess piece at the current state and these moves can be determined.
- Self-Driving Cars- The actions of a self-driving car are not unique, it varies time to time.

3. Competitive vs Collaborative

In a competitive environment, agents compete against each other to optimize their individual outcomes. An agent is said to be in a collaborative environment when multiple agents cooperate to produce the desired output.

Examples:

- Chess - Each agent competes to defeat the opponent and win the match.
- Self-Driving Cars - Vehicles coordinate with each other to ensure safe travel and avoid collisions.

4. Single-agent vs Multi-agent

An environment consisting of only one agent is said to be a single-agent environment. A multi-agent environment involves more than one agent interacting within the same environment.

Examples:

- Maze solving – One agent navigates to find the exit.
- Football – Multiple players from both teams interact to play the game.

5. Dynamic vs Static

A dynamic environment keeps changing while the agent is performing actions or interacting with it. A static environment remains unchanged while the agent is acting on it.

Examples:

- Roller coaster ride – The state keeps changing continuously during motion.
- Empty house – Surroundings do not change when an agent enters.

6. Discrete vs Continuous

If an environment consists of a finite number of actions that can be deliberated in the environment to obtain the output, it is said to be a discrete environment. A continuous environment has uncountable or infinitely varying actions and states.

Examples:

- Chess – It has a limited number of possible moves in any given position.
- Self-driving cars – Actions like steering and acceleration vary continuously and cannot be counted.

7. Episodic vs Sequential

An episodic environment consists of independent tasks where each action is separate and does not depend on previous actions. A sequential environment involves dependent actions where current decisions affect future outcomes.

Examples:

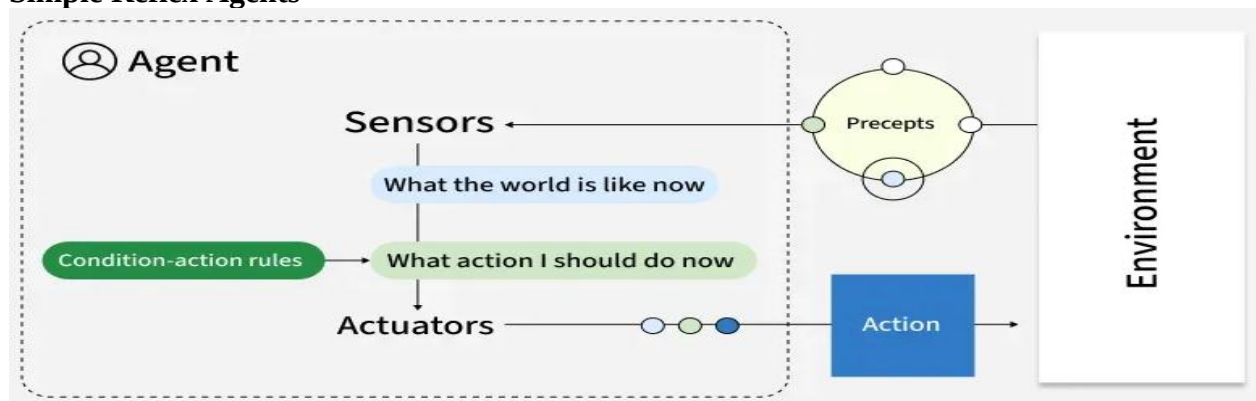
- Pick and place robot – Each item is checked independently on a conveyor belt.
- Checkers – Each move influences all subsequent moves in the game.

8. Known vs Unknown

A known environment is one where the outcomes of all possible actions are already known to the agent. An unknown environment is one where the agent does not know the outcomes and must learn or explore to understand them.

Examples:

- Route planning in a GPS navigation system – Outcomes of paths are already known.
- Exploring an unfamiliar maze – The agent must discover paths and outcomes while navigating.

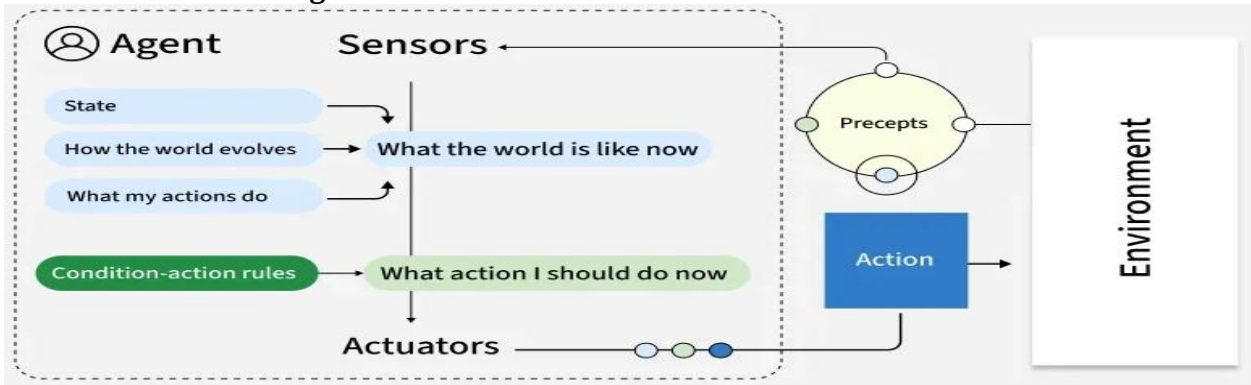
AGENT ARCHITECTURES (SIMPLE REFLEX, MODEL-BASED, GOAL-BASED, UTILITYBASED, LEARNING AGENTS)**1. Simple Reflex Agents**

Simple reflex agents act only on the current input using predefined rules, without memory or history. They are fast and suitable for simple, fully observable environments but perform poorly in dynamic or partially observable situations due to lack of memory and reasoning.

- **Reactive:** These agents respond immediately to inputs without consideration for prior events or predicting future outcomes.
- **Limited Scope:** They excel in predictable environments where tasks are straightforward and the relationships between actions and results are well understood.
- **Quick Response:** Since decisions are made based only on immediate input, it can react without delay.
- **No Learning:** These agents cannot improve or change their behavior based on past experiences.

Example: Traffic light control systems that change signals based on fixed timing.

2. Model-Based Reflex Agents

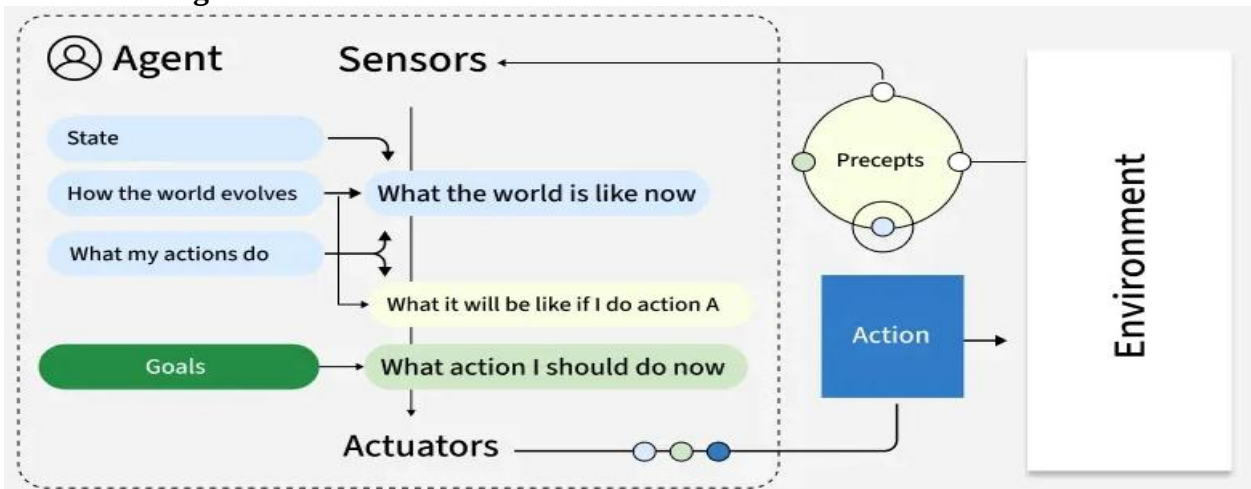


Model-based reflex agents improve simple reflex agents by maintaining an internal state of the environment to handle partially observable and changing situations. However, their decisions are still mainly reactive and depend on the accuracy of the internal model.

- **Internal State:** Maintains an internal model to handle partially observable environments and support flexible decisions.
- **Adaptive:** Updates its internal state based on new information to adjust to changes.
- **Better Decision-Making:** Uses stored information to make more informed and reliable choices.
- **Increased Complexity:** Requires more memory and computation to maintain and update the internal model.

Example: Robot vacuum cleaners that map rooms and tracks cleaned areas.

3. Goal-Based Agents

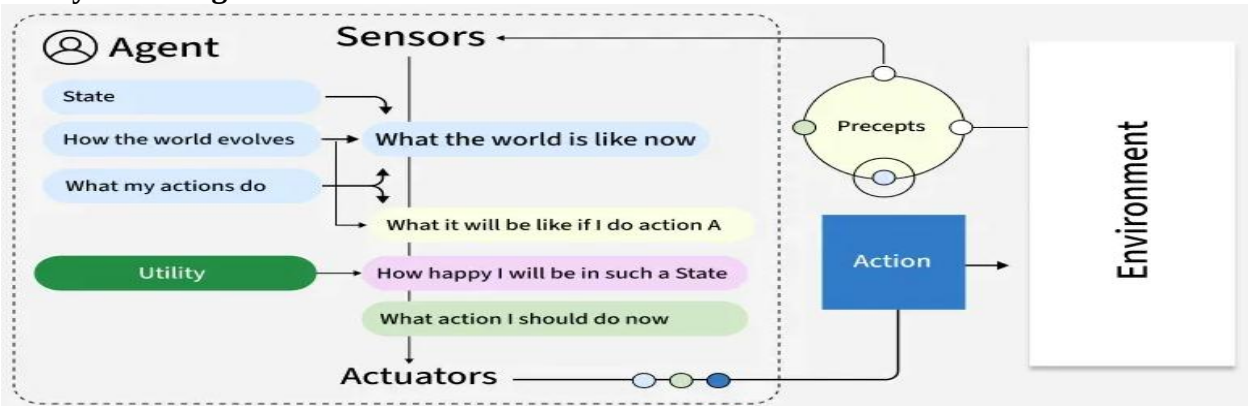


Goal-based agents choose actions by evaluating future states in relation to defined goals. They can plan action sequences to achieve objectives, enabling more flexible problem-solving, but depend on clear goals and effective planning methods in complex environments.

- **Goal-Oriented:** Makes decisions based on how well actions align with defined objectives.
- **Planning and Search:** Uses planning algorithms to evaluate possible action sequences to reach goals.
- **Flexible:** Re-plans and adjusts strategies when conditions or information change.
- **Future-Oriented:** Unlike reflex agents, they think ahead and predict future outcomes to find the best course of action.

Example: Logistics routing agents that find optimal delivery routes based on factors like distance and time. They continuously adjust to reach the most efficient route.

4. Utility-Based Agents

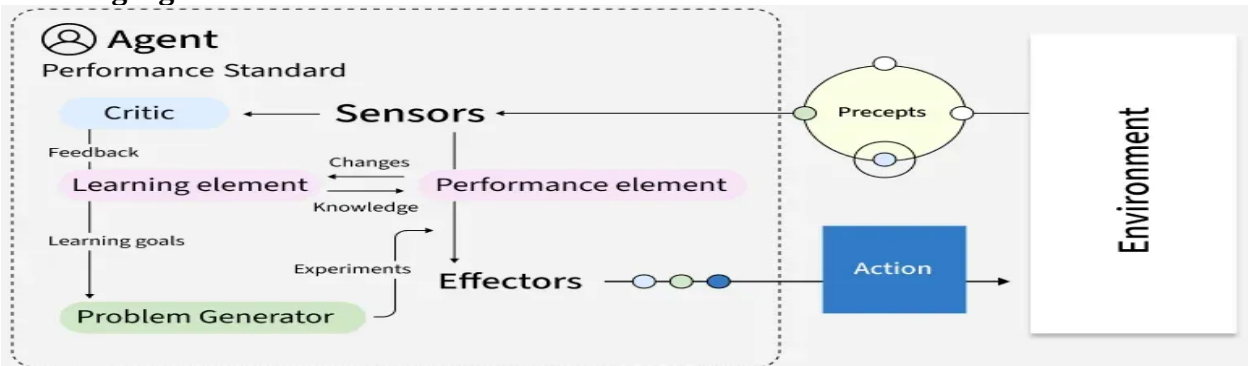


Utility-based agents go beyond goal-based agents by evaluating how desirable different outcomes are using a utility function. This allows them to make trade-offs and choose the most optimal action in uncertain or limited-resource situations. Their performance depends heavily on how well the utility function is designed.

- **Multi-Criteria Decision Making:** Considers multiple factors such as cost, benefit, risk, and time to choose the best action.
- **Trade-Offs:** Make decisions by balancing competing goals and preferences often finding the best "compromise."
- **Subjectivity:** Adapts to user or organizational preferences through customizable utility functions.
- **Increased Complexity:** Finding utility functions for different factors can be computationally intensive and complex.

Example: Financial portfolio management agents that evaluate investments based on factors like risk, return and diversification operate by choosing options that provide the most value.

5. Learning Agents



Learning agents improve performance over time by learning from experience and updating their models or strategies. They adapt to changing environments and typically use supervised, unsupervised, or reinforcement learning.

- **Adaptive Learning:** Improves their decision-making through continuous feedback from their actions.
- **Exploration vs. Exploitation:** Tries balancing new actions and using known successful strategies.
- **Flexibility:** Adapt to a wide variety of tasks or environments by modifying their behavior based on new data.
- **Generalization:** Applies lessons learned in one context to new, similar situations enhancing their versatility.

Example: Customer service chatbots can improve response accuracy over time by learning from previous interactions and adapting to user needs.